

6th Sem/ Chemical (Pulp & Paper)
Subject : Alternate Energy Sources

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 Which of the following is a renewable energy source?
- a) Petroleum b) Coal
c) Solar d) Natural gas
- Q.2 Largest available of wind energy in India is in :
- a) Punjab b) Rajasthan
c) Kerala d) Assam
- Q.3 The process responsible for biomass production is :
- a) Respiration b) Photosynthesis
c) Fermentation d) Transpiration
- Q.4 One major problem associated with OTEC is :
- a) Global warming b) High capital cost
c) Air pollution d) Easy installation
- Q.5 A solar cell works on the principle of :
- a) Conduction b) Convection
c) Photovoltaic effect d) Thermionic emission

- Q.6 Zero energy concept mainly aims at :
- a) Reducing electricity
 - b) Buildings producing as much energy as they consume
 - c) Generating infinite energy
 - d) Nuclear waste reduction

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 State one reason why alternate energy is required worldwide.
- Q.8 Define solar radiation.
- Q.9 Write one major problem associated with wind energy.
- Q.10 Give one example of a bio fuel.
- Q.11 State one limitation of tidal power.
- Q.12 Hydrogen energy is _____.

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 Write short notes on the Future prospects and social implications of alternate energy sources.
- Q.14 Write a short note on the hydrogen fuel cells.

- Q.15 What are applications of solar thermal energy?
- Q.16 Explain the principle of a photovoltaic cell.
- Q.17 Explain the properties of wind energy and its availability in India.
- Q.18 How can urban waste be converted into energy?
- Q.19 Explain the mechanics of tides and waves as energy suppliers.
- Q.20 What are the advantages and limitation of tidal energy.
- Q.21 Compare fossil fuels and alternate energy sources in terms of availability and sustainability.
- Q.22 What are the applications of hydrogen energy and problems associated with hydrogen energy?

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 Write a short note
- a) Photosynthesis
 - b) Bio fuel
 - c) Hydrogen energy
 - d) Wind velocity
- Q.24 Explain in detail the principle, construction and working of any one of Solar thermal power generation with a neat diagram.
- Q.25 Explain the terms
- a) Properties of wind.
 - b) Availability of wind energy in India